Algebra Essentials

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Systems of two equations

Solve the system: $x + y = 10$, $x - y = 4$.

Adding the equations: $2x = 14$, so $x = 7$. Then $y = 10 - 7 = 3$.

2 [Foundational] • Target: Substitution into formulas

The formula for simple interest is $I = Prt$. Find I when $P = 2{,}000$, $r = 0.04$, $t = 3$.

$I = 2{,}000 \times 0.04 \times 3 = 240$.

3 [Foundational] • Target: Substitution into formulas

The perimeter of a rectangle is $P = 2(l + w)$. Find P when $l = 15$ and $w = 9$.

$P = 2(15 + 9) = 2(24) = 48$.

4 [Foundational] • Target: Substitution into formulas

If $v = u + at$, find v when $u = 5$, $a = 3$, $t = 4$.

$v = 5 + 3(4) = 5 + 12 = 17$.

5 [Foundational] • Target: Substitution into formulas

The area of a triangle is $A = \frac{1}{2}bh$. Find A when $b = 14$ and $h = 9$.

$A = \frac{1}{2} \times 14 \times 9 = 63$.

6 [Foundational] • Target: Translating words into equations

Twice a number, plus 7, equals 25. Find the number.

$2x + 7 = 25 \Rightarrow 2x = 18 \Rightarrow x = 9$.

7 [Foundational] • Target: Translating words into equations

A number decreased by 8 is equal to three times the number, decreased by 20. Find the number.

$x - 8 = 3x - 20 \Rightarrow 12 = 2x \Rightarrow x = 6$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Translating words into equations

The sum of three consecutive integers is 84. Find the integers.

Let the integers be n, $n+1$, $n+2$: $3n+3=84 \Rightarrow 3n=81 \Rightarrow n=27$. The integers are $27, 28, 29$.

9 [Intermediate] • Target: Translating words into equations

Five more than three times a number is the same as 10 less than five times the number. Find the number.

$3x+5=5x-10 \Rightarrow 15=2x \Rightarrow x=7.5$.

10 [Intermediate] • Target: Word problems

A rectangle has length $(x+3)$ cm and width x cm. This question has two parts.

a If the area is 40 cm², write an equation in x. Area $= x(x+3)=40$, i.e. $x^2+3x-40=0$.

b Solve for x. Factoring: $(x+8)(x-5)=0$. Since length must be positive, $x=5$.

(a) Area $= x(x+3)=40$, i.e. $x^2+3x-40=0$.; (b) Factoring: $(x+8)(x-5)=0$. Since length must be positive, $x=5$.

11 [Intermediate] • Target: Word problems

A father is 3 times as old as his son. In 12 years, he will be twice as old as his son. Find their current ages.

Let the son's age be x; the father's is $3x$. In 12 years: $3x+12=2(x+12) \Rightarrow 3x+12=2x+24 \Rightarrow x=12$. Son is 12, father is 36.

12 [Intermediate] • Target: Word problems

Tickets cost 25 riyals for adults and 15 riyals for children. A group of 10 people paid 220 riyals in total. How many adults were in the group?

Let a be the number of adults, so $10-a$ are children: $25a+15(10-a)=220 \Rightarrow 25a+150-15a=220 \Rightarrow 10a=70 \Rightarrow a=7$ adults.

13 [Intermediate] • Target: Word problems

The sum of two numbers is 45, and their difference is 11. Find the two numbers.

Let the numbers be x and y with $x+y=45$ and $x-y=11$. Adding: $2x=56 \Rightarrow x=28$, then $y=17$.

14 [Intermediate] • Target: MCQ — Expanding and simplifying

Solve the linear equation for x: $4(2x-3)-3(x+1)=14$.

Step 1: Expand parentheses: $8x-12-3x-3=14 \Rightarrow 5x-15=14$. Step 2: Isolate x: $5x=29 \Rightarrow \mathbf{x=\frac{29}{5}=5.8}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: MCQ — Factoring** • **Cross-Lesson: 7.01 Data Analysis & Probability**

In a survey of 200 students regarding their book reading habits, the data was recorded in a two-way probability table. The following table shows the results of the survey: | | Novels | Science | History | Total | |---------|--------|---------|---------|------| | Boys | 55 | 35 | 10 | 100 | | Girls | 45 | 40 | 15 | 100 | | Total | 100 | 75 | 25 | 200 | If a student is chosen at random from these students, what is the probability that the student is a girl and reads science books?

To calculate the required probability, we need to find the number of girls who read science books and divide this number by the total number of students. From the table, we find that the number of girls who read science books is 40. The total number of students is 200. Probability = (Number of girls who read science books) / (Total number of students) Probability = 40 / 200 Probability = 4 / 20 Probability = 1 / 5 Probability = 0.200

16 **[Challenging]** • **Target: MCQ — Solving quadratic equations** • **Cross-Lesson: 8.01 Word Problems & Reasoning**

If $\frac{1}{x}+\frac{1}{y}=\frac{1}{3}$ and $x+y=8$, find the value of the product xy.

Step 1: Combine the reciprocals over a common denominator: $\frac{x+y}{xy}=\frac{1}{3}$. Step 2: Substitute $x+y=8$ and cross-multiply: $\frac{8}{xy}=\frac{1}{3}\Rightarrow xy=8\times 3=\mathbf{24}$.

17 **[Challenging]** • **Target: MCQ — Systems of two equations** • **Cross-Lesson: 1.01 Arithmetic & Number Sense**

Pipe A fills a swimming pool in 4 hours, and Pipe B fills it in 6 hours. If both pipes are opened simultaneously, in how many hours will the pool be full?

Step 1: Use combined rate formula $T=\frac{t_1t_2}{t_1+t_2}$: $T=\frac{4\times 6}{4+6}=\frac{24}{10}=\mathbf{2.4}$ hours (2 hours 24 minutes).

18 **[Challenging]** • **Target: MCQ — Laws of exponents** • **Cross-Lesson: 2.01 Ratios, Proportions & Percentages**

A train travels from Riyadh to Dammam at 100 km/h and returns at 150 km/h. What is the average speed of the train for the complete round trip?

Step 1: Apply harmonic mean for identical round-trip distance:
$s_{\text{avg}}=\frac{2s_1s_2}{s_1+s_2}=\frac{2(100)(150)}{100+150}=\frac{30000}{250}=\mathbf{120}$ km/h.

19 **[Challenging]** • **Target: MCQ — Inequalities** • **Cross-Lesson: 4.01 Geometry & Measurement**

A father's age is three times his son's age. If the sum of their ages is 48 years, how old is the son?

Step 1: Set up algebraic equation: Let son's age be x. Father's age is $3x$. $x+3x=48\Rightarrow 4x=48$ Step 2: Solve for x: $x=\frac{48}{4}=\mathbf{12}$ years.

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20 [Challenging] • Target: MCQ — Reading function graphs • Cross-Lesson: 5.01 Quantitative Comparisons

A car travels 240 km in 3 hours. At this same constant speed, how many kilometers will it travel in 5 hours?

Step 1: Compute unit speed: $s = \frac{240 \text{ km}}{3 \text{ h}} = 80$ km/h. Step 2: Multiply by new time: $d = 80 \times 5 = \mathbf{400}$ km.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Solving quadratic equations

The graph shows $y = x^2 - 49$. What are its roots (where the curve crosses the x-axis)?

a. x = 7 only

b. x = 49

c. **x = 7 or x = -7 ✓**

d. x = -7 only

The curve crosses the x-axis at $x = -7$ and $x = 7$, matching $x^2 - 49 = 0$.

22 [Foundational] • Target: MCQ — Systems of two equations

The graph shows two lines. What is their point of intersection?

a. (1, 4)

b. (0, 5)

c. **(2, 3) ✓**

d. (3, 2)

The two lines $y = x + 1$ and $y = -x + 5$ cross where both are true: at $(2, 3)$.

23 [Foundational] • Target: MCQ — Inequalities

The number line shows the solution set of an inequality. Which inequality matches?

a. $x \geq 4$

b. $x \leq 4$

c. $x > 4$

d. $x < 4$ ✓

An open circle at 4 with shading to the left means $x < 4$.

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24 [Foundational] • Target: MCQ — Reading function graphs

The graph shows a line. What is its slope?

a. $\frac{1}{2}$

b. 2 ✓

c. -2

d. -1

The line rises 2 for every 1 across, e.g. from $(0, -1)$ to $(1, 1)$: slope $= 2$.

25 [Foundational] • Target: MCQ — Reading function graphs

What is the y-intercept of the line shown?

a. 4 ✓

b. 0

c. -4

d. 1

The line crosses the y-axis at $(0, 4)$.

26 [Foundational] • Target: MCQ — Reading function graphs

What is the equation of the line shown?

a. $y = 3x + 1$

b. $y = x + 3$

c. $y = \frac{1}{3}x$

d. $y = 3x$ ✓

The line passes through the origin and $(1, 3)$: slope 3, no intercept, so $y = 3x$.

27 [Foundational] • Target: MCQ — Reading function graphs

Find the slope of the line through the two marked points.

a. 2 ✓

b. $\frac{1}{2}$

c. -2

d. 4

Slope $= \frac{6-2}{3-1} = \frac{4}{2} = 2$.

28 [Foundational] • Target: MCQ — Substitution into formulas

Given $A = \frac{1}{2}bh$, find A when $b = 10$ and $h = 6$.

a. 16

b. 8

c. 60

d. 30 ✓

$A = \frac{1}{2}(10)(6) = 30$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Substitution into formulas**

The rectangle shown has area $A = lw = 48$ and length $l = 8$. Find the width w.

a. 8

b. 6 ✓

c. 384

d. 40

$48 = 8w \Rightarrow w = 6$.

30 **[Intermediate]** • **Target: MCQ — Translating words into equations**

Three times a number, minus 5, equals twice the number plus 7. Find the number.

a. 19

b. 2

c. -12

d. 12 ✓

$3n - 5 = 2n + 7 \Rightarrow n = 12$.

31 **[Intermediate]** • **Target: MCQ — Translating words into equations**

The sum of two consecutive even integers is 54. Find the larger integer.

a. 28 ✓

b. 27

c. 29

d. 26

$n + (n + 2) = 54 \Rightarrow n = 26$; larger integer $= 28$.

32 **[Intermediate]** • **Target: MCQ — Translating words into equations**

Five less than three times a number is 25. Find the number.

a. 10 ✓

b. 20

c. 5

d. 15

$3n - 5 = 25 \Rightarrow n = 10$.

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33 [Intermediate] • Target: MCQ — Translating words into equations

The perimeter of a rectangle is 54 cm. The length is 3 cm more than the width. Find the width.

a. 24

b. 12 ✓

c. 15

d. 9

$2(w + (w + 3)) = 54 \Rightarrow 4w + 6 = 54 \Rightarrow w = 12$.

34 [Intermediate] • Target: MCQ — Word problems

A number increased by 20% equals 72. Find the number.

a. 90

b. 52

c. 60 ✓

d. 86.4

$1.2n = 72 \Rightarrow n = 60$.

35 [Intermediate] • Target: MCQ — Word problems

Ahmed is twice as old as his sister. In 5 years, the sum of their ages will be 40. Find Ahmed's current age.

a. 20 ✓

b. 10

c. 15

d. 25

Let sister $= x$, Ahmed $= 2x$. $(x + 5) + (2x + 5) = 40 \Rightarrow 3x = 30 \Rightarrow x = 10$; Ahmed $= 20$.

36 [Intermediate] • Target: MCQ — Word problems

A positive number plus its square is 30. Find the number.

a. 30

b. -6

c. 6

d. 5 ✓

$n^2 + n - 30 = 0 \Rightarrow (n + 6)(n - 5) = 0$; the positive root is $n = 5$.

37 [Intermediate] • Target: MCQ — Word problems

The bar chart shows four students' test scores. Whose score is exactly 12 more than 70?

a. Student B

b. Student D

c. **Student C** ✓

d. Student A

$70 + 12 = 82$, which is Student C's score.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

The product of two consecutive integers is 132. Find the larger integer.

a. 13

b. 12 ✓

c. 10

d. 11

$n(n+1)=132 \Rightarrow n^2+n-132=0 \Rightarrow (n+12)(n-11)=0$; taking the positive root $n=11$, larger $=12$.

39 [Challenging] • Target: MCQ — Inequalities • Cross-Lesson: 4.01 Geometry & Measurement

■■■ ■■■ $f(x)=x^2-4x+3$ ■ $g(x)=x-1$■ ■■■■■ ■■■■ ■■■■■ ■■■■■ ■■■■■■■ ■■■■■■■■ $f(x)$ ■ $g(x)$ ■■■■■ $x=1$. If $f(x)=x^2-4x+3$ and $g(x)=x-1$, find the point of intersection of the graphs of $f(x)$ and $g(x)$ when $x=1$.

a. **(1, 0)** ✓

b. (0, 1)

c. (1, 1)

d. (0, 0)

■■■■■■ ■■■■ ■■■■■■■■■ ■■■■■ ■■■ ■■■■■ ■■■■ $f(x)$ ■ $g(x)$ ■■■ $x=1$. ■■■■■■■■ ■■■■■■ $f(x)=x^2-4x+3$: $f(1)=(1)^2-4(1)+3=1-4+3=0$. ■■■■■■■■ ■■■■■■ $g(x)=x-1$: $g(1)=1-1=0$. ■■■■ ■■■■ ■■■■■■■■ ■■■ $x=1$ ■■ $(1,0)$. To find the point of intersection, we need to find the value of $f(x)$ and $g(x)$ at $x=1$. For the function $f(x)=x^2-4x+3$: $f(1)=(1)^2-4(1)+3=1-4+3=0$. For the function $g(x)=x-1$: $g(1)=1-1=0$. Therefore, the point of intersection at $x=1$ is.

40 [Challenging] • Target: MCQ — Reading function graphs • Cross-Lesson: 5.01 Quantitative Comparisons

In the given function graph, what are the coordinates of the y-intercept of the parabola $y=x^2-4x+3$?

a. **(0, 3)** ✓

b. (3, 0)

c. (1, 0)

d. (0, -3)

The y-intercept of a function occurs when $x=0$. Substituting $x=0$ into the equation $y=x^2-4x+3$, we get: $y=(0)^2-4(0)+3$ $y=0-0+3$ $y=3$ Therefore, the coordinates of the y-intercept are $(0,3)$. The point $(1,0)$ is shown on the graph as an x-intercept. $y=(0)^2-4(0)+3$ $y=0-0+3$ $y=3$ ■■■■■ ■■■■■■■■ ■■■■ ■■■■■ ■■■■■■ ■■■■■■ ■■ $(0,3)$. ■■■■■■ $(1,0)$ ■■■■■ ■■■ ■■■■■ ■■■■■■■ ■■■■■ ■■■■■ ■■ ■■■■■■ ■■■■■■ (x-intercept).

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41 [Challenging] • Target: MCQ — Substitution into formulas • Cross-Lesson: 6.01 Sequences & Patterns

■■■■■■ $f(x) = 2x + 3$■ ■■ ■■ ■■■■ $f(5)$■ For the function $f(x) = 2x + 3$, what is the value of $f(5)$?

a. 8

b. 10

c. **13** ✓

d. 23

■■■■■■ ■■■■ $f(5)$■ ■■■■ ■■ x ■■ 5 ■■ ■■■■■ ■■■■■■ $f(x) = 2x + 3$. $f(5) = 2(5) + 3$ $f(5) = 10 + 3$ $f(5) = 13$ ■■■■■ ■■■■ $f(5)$ ■■ 13. To find the value of $f(5)$, we substitute x with 5 in the function definition $f(x) = 2x + 3$. $f(5) = 2(5) + 3$ $f(5) = 10 + 3$ $f(5) = 13$ Therefore, the value of $f(5)$ is 13.

42 [Challenging] • Target: MCQ — Translating words into equations • Cross-Lesson: 7.01 Data Analysis & Probability

■■■ ■■■■■ ■■■■■■ ■■■■■■ ■■■■■ ■■■■■■■ ■■■■■■ ■■■ ■■ ■■■■■ ■■■ ■■■■ ■■■■■■ ■■■■■ ■■■■■■■■ ■■■■■■■■. ■■■■ ■■■ ■■■■■ ■■■■■ 36 ■■■■■ ■■■■■ 2 ■■■. ■■■■ ■■■ ■■■■■ ■■■■■■ 48 ■■■■■ ■■■■■ 2 ■■■. ■■■ ■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■■■■ ■■ 25 ■■■■■■■ ■■■ ■■ ■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■ ■■■■■■■■ ■■ ■■■■■ ■■ ■■■■■■■■ ■■ ■■■■■■■ ■■■■■■ ■■■■ ■■■■■■ ■■■ ■■■■■ ■■■■ An architect designed a rectangular garden. A portion of this garden was designated for two perpendicular straight pathways. The first pathway is 36 meters long and 2 meters wide. The second pathway is 48 meters long and 2 meters wide. If the cost of paving one square meter is 25 riyals, what is the minimum possible cost to pave both pathways, considering that their intersection will be paved only once?

a. 3000 ■■■■

b. 3100 ■■■■

c. **3200 ■■■■** ✓

d. 3300 ■■■■

■■■■■ ■■■ ■■■■■ ■■■■■ ■■■■■■ ■■■ ■■■■■■■■ ■■■ ■■■■■ ■■■■ ■■■■■■■ ■■■■■■ ■■■■■■■ ■■ ■■■■ ■■■■ ■■■■■ ■■■■■■■■ ■■■■■■. ■■■■■ ■■■■■ ■■■■■ = ■■■■■ × ■■■■■ = 36 ■■■ × 2 ■■■ = 72 ■■■ ■■■■. ■■■■■ ■■■■■ ■■■■■■ = ■■■■■ × ■■■■■ = 48 ■■■ × 2 ■■■ = 96 ■■■ ■■■■. ■■■■■ ■■■■■■■ ■■ ■■■■■■ ■■■■■■■ ■■■ ■■■■■■■. ■■■■■ ■■■ ■■■ ■■■ ■■■■■■■ ■■ 2 ■■■■ ■■■ ■■■■■ ■■■■■ ■■■■■■■ ■■: 2 ■■■ × 2 ■■■ = 4 ■■■ ■■■■. ■■■■■ ■■■■■■■ ■■■■■■ ■■■■ ■■■■ ■■■■■■■■ ■■■■ ■■■■■■ ■■■■■■■ ■■ ■■■■ ■■■■■ ■■■■■■■ ■■■■ ■■ ■■■■■■ ■■■■■: ■■■■■■■ ■■■■■■ = ■■■■■ ■■■■■ ■■■.

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43 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 8.01 Word Problems & Reasoning

[illegible]. [illegible] 1200 [illegible] 30 [illegible]. [illegible] 1800 [illegible] 45 [illegible]. [illegible] [illegible] [illegible]

45 [Challenging] • Target: Solving linear equations • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

[illegible] [illegible] [illegible] $f(x) = ax^2 + bx + c$. [illegible] [illegible] $f(1) = 5$[illegible] $f(2) = 11$[illegible] [illegible] $f(3) = 19$[illegible] [illegible] [illegible] [illegible] $a + b + c$[illegible] [illegible] [illegible] a[illegible] b[illegible] [illegible] c [illegible] [illegible] Let the function be $f(x) = ax^2 + bx + c$. If $f(1) = 5$, $f(2) = 11$, and $f(3) = 19$, what is the value of $a + b + c$, given that a, b, and c are real constants?

a. **5** ✓

b. 7

c. 10

d. 12

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: 1. $f(1) = a(1)^2 + b(1) + c = a + b + c = 5$ 2. $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = 11$ 3. $f(3) = a(3)^2 + b(3) + c = 9a + 3b + c = 19$ [illegible] [illegible] [illegible] [illegible] $a + b + c$[illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] $f(1)$ [illegible] $a + b + c$[illegible] [illegible] [illegible] [illegible] 5. We have the following equations from the given conditions: 1. $f(1) = a(1)^2 + b(1) + c = a + b + c = 5$ 2. $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = 11$ 3. $f(3) = a(3)^2 + b(3) + c = 9a + 3b + c = 19$ We are asked to find t.

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